

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of MSc (Microbiology) Examination (Backlog) March 2018

Course code & Name: MI 815 Agricultural Microbiology

Date: 30/03/2018 Day: Friday Time: 1:30 pm To 1:50 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
-

Q - I Choose the correct answer for the following questions.

10

1. diazotrophs are_____.
(i) Organisms capable of nitrogen fixation.
(ii) Organisms such as *rhizobia*, *azotobacter*, *azospirillum*.
(iii) Organisms capable of phosphorus solubilization.
(iv) (i) and (ii) both
2. *Beauveria bassiana* is an insecticidal fungi that controls insects by _____.
(i) Enzymatic degradation of insect cuticle. (ii) Secretion of antibiotic.
(iii) Releasing Unregulated waste products. (iv) All of these
3. Plant absorbs N_2 in the form of _____.
(i) Nitrite (NO_2^-) (ii) Ammonium (NH_4^+)
(iii) Nitrate (NO_3^-) (iv) All of these

4. The characteristics feature of VAM is it penetrates plant cell wall and form____.
- (i) Spores intracellularly.
 - (ii) Haustoria.
 - (iii) Vesicles and dichotomously branched invaginations called arbuscules.
 - (iv) Massive spore forming structures intracellularly.
5. The main precursor for IAA biosynthesis pathways in bacteria is_____.
- (i) Tyrosine
 - (ii) Tryptophan
 - (iii) Tryptase
 - (iv) None
6. Which of the following is used as carrier in biofertilizer formulations?
- (i) Peat
 - (ii) Talc
 - (iii) Vermiculite
 - (iv) All of these
7. What is the first step of soil formation?
- (i) Activity of microbes
 - (ii) Activity of plants
 - (iii) Weathering of rocks
 - (iv) All of these
8. Which of the following molecular techniques generally used for the identification of rhizospheric microorganisms.
- (i) 16s RNA sequencing
 - (ii) 18s RNA sequencing
 - (iii) 2-D gel electrophoresis
 - (iv) (i) and (ii) both
9. Which of the following compounds are produced by both plants and microbes?
- (i) Siderophores
 - (ii) IAA
 - (iii) DAPG
 - (iv) both (i) and (ii)
10. Fungi, which can grow on living host plant, are called_____.
- (i) Obligate saprophyte
 - (ii) Obligate parasite
 - (iii) Facultative parasite
 - (iv) Saprophytes

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of MSc (Microbiology) Examination (Backlog) March 2018

Course code & Name: MI 815 Agricultural Microbiology

Date: 30/03/2018 Day: Friday Time: 1:51 pm To 3:30 pm Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION – I

Q – II Answer the following questions as directed (any five) (2 Marks X 5) 10

1. Write the differences between Endo-mycorrhiza and Ecto-mycorrhiza.
2. What is R: S ratio? Explain in brief with example.
3. Explain “Trojan horse” mechanism of killing nematodes by bacteria.
4. Which is the causative organism of crown gall disease? Explain the steps involved in crown gall formation.
5. When a pot with plants is inoculated with high IAA producing *pseudomonas* strain, the plants showed stunted root growth. What might be the possible reasons? How it can be reduced/rectified?
6. Write a note on “microorganisms associated with organic matter decomposition”
7. Draw the labeled diagram of nodule formation in leguminous plant by *Rhizobium*.

SECTION – II

Q-III Answer the following questions as directed (any three) (5 Marks X 3) 15

1. What is biofertilizer? Which are the types of Biofertilizer? Explain the steps for large scale production of biofertilizer.
2. What is Rhizosphere? Explain in detail about the isolation and identification of rhizospheric microorganisms.
3. Write any three major fungal plant diseases, causative organisms and their hosts.
4. What is biogeochemical cycle? Explain in brief about nitrogen and phosphorus cycle.